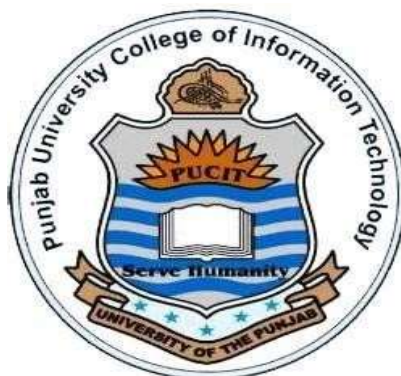




Punjab University College Of Information Technology

Final Year Project Proposal

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PROJECT TITLE	AI-Enhanced Admission Management System
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STUDENT INFORMATION				
Sr.	Student ID	Name	Email	Mobile #
1.	081127	Rida Nadeem	rforridanadeem03@gmail.com	0326-2651860
2.	081119	Muhammad Ahmad	Muah327@gmail.com	0370-0140162
3.	081122	Hafiz Muhammad Awais	Mawaisrajpoot03@gmail.com	0316-4011101

PROBLEM STATEMENT

The university admission process in Pakistan is largely manual, inefficient, and error-prone. Students fill forms manually, often entering incorrect or fake information. They apply to ineligible programs, wasting both time and administrative resources. Merit list generation is slow, lacks transparency, and can be perceived as biased.

Administrators struggle with document verification, application management, student categorization, and reporting. There is no clear visualization of how many students are admitted into each category (merit, quota, self-finance), nor insights into percentage-based eligibility trends.

A modern solution is required: a data-driven, AI-enhanced admission system that automates the entire cycle (from pre-application form filling & data extraction to admin-level analytics & insights) ensuring transparency, fairness, and efficiency.

EXECUTIVE SUMMARY

Our proposed AI-Enhanced Admission Management System streamlines admissions by:

- Automating Pre-Application with OCR-based CNIC & certificate data extraction.
- Ensuring Transparency via eligibility checks and automated merit list generation.
- Recommending Programs using AI-driven logic based on student marks and subjects.
- Empowering Administrators with dashboards, data visualization, and category-based analytics.
- Providing Insights into admission trends, percentage distributions, and subject-specific performance without exposing personal data.

This system reduces manual workload, improves accuracy, enhances student experience, and provides administrators with data-driven decision-making.

INTRODUCTION

Admissions are the foundation of academic institutions, yet most universities in Pakistan rely on outdated systems or manual methods.

Students face confusion in program selection, delays in merit lists, and lack of clarity on eligibility.

Administrators deal with thousands of applications, tedious verification, and manual merit preparation.

This project proposes an AI-powered Admission Management System that introduces automation, transparency, and smart decision support. It integrates document verification, program recommendation, automated merit lists, student categorization, and data visualization into a single unified platform.

COMPETITOR ANALYSIS

Existing admission portals provide basic online application and fee payment but lack AI-based recommendations, merit transparency, and admin analytics.

Current ERP solutions are expensive, complex, and not tailored to student/administrator needs.

Our system is unique because it offers:

- OCR-based pre-application automation.
- Eligibility verification & program recommendation using AI.
- Automated merit list generation.
- Admin dashboards with insights & student categorization.

OBJECTIVES

- **Automate Form Filling & Data Extraction** – Use OCR to extract information from CNICs and academic documents.
- **Eligibility & Merit List Generation** – Ensure accurate eligibility checks and transparent merit list creation.
- **AI-Based Program Recommendation** – Suggest suitable programs based on student academic profile and preferences.
- **Admin Dashboard & Visualization** – Provide analytics on admissions by category (merit, quota, self-finance) with clear percentage-based distribution.
- **Fairness, Security & Scalability** – Maintain unbiased admissions, ensure data protection, and support system growth.

MOTIVATION

The admission process in many educational institutions is often challenging for both students and administrators. Students frequently lack proper guidance and face limited transparency during the process, which can lead to confusion and misinformed decisions. On the other hand, administrators struggle with repetitive tasks and managing large-scale admission data, making the process time-consuming and error-prone. By introducing an AI-enhanced admission management system, bias and human errors can be minimized, ensuring a fair, transparent, and efficient process. Furthermore, integrating AI into education systems aligns with modern research and industry trends, making this project both academically valuable and practically relevant. A centralized digital platform will help streamline the entire admission cycle, reducing delays and miscommunication while improving the overall user experience. Finally, the project aims to provide a scalable prototype that universities can adopt, adapt, and expand for real-world implementation.

REQUIREMENTS

Functional Requirements

❖ Student Side:

- CNIC & document upload with OCR-based data extraction.
- Eligibility check with percentage & subject-based logic.
- AI-based program recommendation.
- Program priority selection.
- Real-time application tracking.
- Merit list notifications.

❖ Admin Side:

- Dashboard with visual analytics.
- Categorization of students (Merit / Quota / Self-Finance).
- Subject-wise & percentage-based insights.
- Report export & monitoring tools.

❖ System Side:

- Automated merit list generation.
- Fee management (forms, defaulter tracking).
- Secure authentication and access control.
- Scalability for thousands of applications.

Non-Functional Requirements

- **Performance:** Real-time processing for thousands of applications.
- **Scalability:** Cloud-hosted system for peak loads.
- **Security:** Data encryption, 2FA login, secure payments (sandbox).
- **Reliability:** 99% uptime with daily backups.
- **Usability:** Mobile-friendly, bilingual interface (English/Urdu).

FEATURES OF PROJECT

Student Features

- **Document Upload & OCR** – Upload CNIC and academic certificates with automatic data extraction.
- **Eligibility Check** – Automatic verification of program requirements.
- **AI-Based Program Recommendation** – Suggests suitable programs based on student profile and marks.
- **Online Aptitude/Quiz Test** – Assess student skills and subject knowledge.
- **Program Priority Selection** – Students can rank and submit their preferred programs.
- **Application Tracking** – Real-time status updates of submitted applications.
- **Merit List Notification** – Instant alerts for admission results and updates.

Admin Features

- **Dashboard & Analytics** – Visualize applications, admissions by category (merit/quota/self-finance), and percentage-based distributions.
- **Merit List Automation** – Transparent, rule-based merit list generation with AI support for edge cases.
- **Student Categorization** – Automatic grouping into merit, quota, and self-finance categories.

- **Performance Insights** – Subject-wise performance and pass/fail percentage analysis.

. System Features

- **Secure Data Management** – Encryption, role-based access, and secure backups.
- **Scalable Deployment** – System designed to handle large numbers of users.
- **User-Friendly Interface** – Accessible on both mobile and desktop devices.
- **Transparency & Fairness** – Clear, explainable AI logic to minimize bias.

ARCHITECTURAL DESIGN

Hardware:

- **Student Devices:** Mobile, PC, Tablets.
- **Server:** Cloud-hosted with load balancing.

Software Stack:

- **Frontend:** React.js
- **Backend:** Python
- **Database:** Mongo DB
- **AI/ML:** OCR (Tesseract), Recommendation models (scikit-learn).
- **Visualization:** Power BI
- **Deployment:** AWS

IMPLEMENTATION METHODOLOGY

The project will be developed using the Agile Scrum Methodology, ensuring flexibility, iterative progress, and continuous feedback.

- **Agile Development (Scrum-based)** – The project will be divided into short sprints, each focusing on specific deliverables.
- **Prototyping of Key Features** – Early prototypes will be created for core modules such as OCR-based document verification, Merit List generation, and the Admin Dashboard to gather feedback and refine usability.
- **Iterative Testing & Deployment** – Each sprint will include testing for functionality, performance, and usability, followed by deployment on a cloud platform for scalability.

Development Phases

- **Requirement Gathering & Design** – Define user roles, workflows, and system architecture with wireframes and mockups.
- **Pre-Application Module** – Implement OCR-based document upload, data extraction, and eligibility checks.
- **Program Recommendation & Merit List** – Develop AI-based recommendation logic and automate merit list generation.
- **Admin Dashboard & Analytics** – Build visualization tools to show admissions by category, merit distribution, and performance insights.
- **Testing & Feedback** – Conduct functional, usability, and security testing; refine based on feedback from students and administrators.
- **Deployment & Documentation** – Deploy the final system on a scalable cloud environment and prepare technical documentation for maintainability.

PROJECT PLAN

Division of Responsibilities

This project will be carried out collaboratively by Rida Nadeem, Muhammad Ahmad and Hafiz Awais.

- **Rida Nadeem** – Frontend development (student & admin dashboards, UI/UX), plus integration of selected AI modules (OCR, Recommendation).
- **Hafiz Awais** – Backend development, database design, API integration, authentication, and cloud deployment.
- **Muhammad Ahmad** – System architecture, testing, documentation, and implementation of AI logic for eligibility checks and merit list.

All members will work collaboratively to ensure smooth progress and successful completion.

Timeline with Milestones (Tentative)

Phase 1: Requirement Analysis & Planning (Weeks 1–2)

Milestone 1: Finalized Requirements & Project Plan

- **Task 1.1:** Literature review on AI in admissions + competitor analysis (Team – Collaborative)
- **Task 1.2:** Finalize technology stack & tools (Team – Collaborative)
- **Task 1.3:** Define use cases, functional & non-functional requirements (Team – Collaborative)
- **Task 1.4:** Create detailed Gantt chart & initial project setup (Team – Collaborative)

Phase 2: System Design (Weeks 3–4)

Milestone 2: Architecture & UI/UX Ready

- **Task 2.1:** Wireframing & mockups for student/admin dashboards (Lead: Rida Nadeem)
- **Task 2.2:** Database schema design (Lead: Hafiz Awais)
- **Task 2.3:** High-level system architecture diagram (Team – Collaborative)
- **Task 2.4:** AI workflow design (OCR, recommendation, merit logic) (Lead: Muhammad Ahmad + Support)

Phase 3: Core Backend Development (Weeks 5–7)

Milestone 3: Backend APIs Functional

- **Task 3.1:** User authentication & access control (Hafiz Awais)
- **Task 3.2:** Application form + OCR-based document upload (Rida Nadeem + Muhammad Ahmad)
- **Task 3.3:** Eligibility check logic (Muhammad Ahmad)
- **Task 3.4:** Merit list automation API (Hafiz Awais + Muhammad Ahmad)

Phase 4: Frontend Development (Weeks 8–9)

Milestone 4: Dashboards Functional

- **Task 4.1:** Student dashboard (application tracking, priority selection, merit list view) (Rida Nadeem)
- **Task 4.2:** Admin dashboard (applications overview, analytics charts) (Rida Nadeem)
- **Task 4.3:** Integration of backend APIs with frontend (Rida Nadeem + Hafiz Awais)
- **Task 4.4:** UI enhancements & responsiveness (Rida Nadeem)

Phase 5: Testing & Optimization (Week 10)

Milestone 5: Tested & Optimized System

- **Task 5.1:** Unit testing (backend APIs, AI modules) (Muhammad Ahmad)
- **Task 5.2:** Unit testing (frontend dashboards) (Rida Nadeem)
- **Task 5.3:** Integration testing (full system) (Team – Collaborative)
- **Task 5.4:** Bug fixing & performance optimization (Team – Collaborative)

Phase 6: Deployment & Documentation (Weeks 11–12)

Milestone 6: Final Deployment & Submission

- **Task 6.1:** Deploy system on cloud (AWS/Azure/GCP) (Hafiz Awais)
- **Task 6.2:** Security setup (SSL/TLS, role-based access, backups) (Hafiz Awais)
- **Task 6.3:** Prepare technical documentation & user manual (Muhammad Ahmad)
- **Task 6.4:** Final project submission & presentation (Team – Collaborative)

EXPECTED OUTCOMES

- Automated pre-application & eligibility checking.
- Transparent and faster merit list generation.
- Smart program recommendation system.
- Admin-level analytics (students per program, merit vs quota distribution, subject-wise insights).
- Secure, scalable, and reliable admission platform.

CONCLUSION

The AI-Enhanced Admission Management System provides a complete admission cycle solution—from form filling and eligibility to analytics and visualization. It ensures fairness, transparency, and efficiency, reducing workload while improving decision-making for administrators.

VERSION CONTROL

1. 02-10-2025 Initial Draft based on project details
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